

Needs Expression / Problem Statement

AI-Based Intrusion Detection System

Detection of *Port Scan* & *Network Reconnaissance* Attacks

Module: Artificial Intelligence — Semester 2

Team: 5 members (Engineering — ENSA)

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1. Context and Current Situation

In the field of network security, the early detection of reconnaissance activities represents a critical challenge. *Port Scanning* — the technique by which an attacker systematically probes the ports of a network to identify active hosts, exposed services, and exploitable vulnerabilities — constitutes the first phase of virtually all cyberattacks documented under the *Cyber Kill Chain* framework.

Currently, network infrastructures rely primarily on static rule-based firewalls and signature-based Intrusion Detection Systems (IDS) such as Snort and Suricata. These tools operate through known pattern matching and apply fixed thresholds (e.g., “more than 100 SYN connections within 10 seconds”).

This approach gives rise to several major shortcomings:

- **Blindness to slow and stealthy scans:** attackers deliberately space their probes (1 packet/second or less) to remain below detection thresholds, rendering static rules ineffective.
- **High false positive rate:** legitimate traffic (software updates, service discovery, monitoring scans) frequently triggers the same signatures as malicious scans, generating a volume of irrelevant alerts that exhausts network administrators.
- **Lack of behavioral profiling:** current systems treat each packet in isolation, without analyzing the temporal and statistical patterns of traffic per source IP over sliding time windows.
- **Inability to classify scan type:** traditional IDS cannot distinguish a *SYN Stealth Scan* from a *UDP Scan*, an *ACK Scan*, or an *XMAS Scan*, depriving the administrator of essential tactical information.

These limitations directly impact Security Operations Center (SOC) teams in terms of response time, alert reliability, and the ability to contain an attack before it progresses to more destructive phases (exploitation, lateral movement, exfiltration).

2. Problem Statement and Needs Expression

Given these observations, it becomes necessary to rethink the detection mechanism in order to **overcome the limitations of static signature-based approaches** and equip the network with an intelligent, adaptive, real-time analytical capability.

The primary objective is to design a system capable of automatically learning the behavioral patterns of network traffic — both normal and malicious — in order to autonomously detect reconnaissance activities, including slow and stealthy variants.

In this context, several questions arise:

1. Which network traffic *features* effectively discriminate a port scan from legitimate behavior?
2. How can a machine learning model detect scans distributed over time, where static thresholds fail?
3. What labeled and representative data must be provided to the model to ensure reliable classification with a minimal false positive rate?

The problem statement can thus be formulated as follows:

*“How can we design, train, and evaluate a Machine Learning model capable of classifying network traffic with high reliability to detect **Port Scan & Reconnaissance** attacks, while drastically minimizing the false positive rate within a controlled network infrastructure?”*

This problem statement aims to deliver a solution that enables: reduction of detection time, improvement of alert accuracy, and decision support for network administrators.

3. Proposed Solution

To address this problem, an approach based on **artificial intelligence**, and more specifically on **supervised classification learning**, can be adopted.

The system will exploit labeled network traffic data sourced from recognized IDS benchmark datasets. By leveraging features such as the number of distinct ports probed per IP, the ratio of SYN packets without completed handshakes, flow duration, and the entropy of the targeted port range, the model will learn to distinguish benign flows from reconnaissance flows.

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| Inputs: | Feature vectors extracted from network traffic per time window (10s and 60s) and per source IP: TCP flag counters, distinct port count, inter-arrival time, port entropy, flow duration. |
| Outputs: | Binary classification (BENIGN / PORT_SCAN) with confidence score; extended to multi-class scan type classification (SYN / UDP / ACK / XMAS / Connect). |
| Model type: | Supervised classification (ensemble decision trees) complemented by an unsupervised anomaly detection layer for slow scans. |

4. Two Proposed AI Solutions

4.1. Automation-Oriented Solution

A **fully autonomous** system where the classification model, once deployed, analyzes traffic in real time, issues alerts, and **automatically applies** countermeasures (blocking the source IP via **iptables**, redirecting to a *honeypot*). No human intervention is required between detection and response.

- *Advantage:* minimal response time (< 15s from detection to blocking).
- *Risk:* a false positive results in blocking a legitimate user without human validation.

4.2. Decision-Support-Oriented Solution

A system that analyzes traffic and **presents results on an interactive dashboard** (port heat map, attacker profile, alert feed with confidence score and severity level). The administrator **manually decides** whether to block the suspicious IP.

- *Advantage:* full human control; risk of blocking due to false positives is eliminated.
- *Risk:* response time depends on administrator availability.

5. Data Dictionary

5.1. Data Sources

Dataset	Subset	Approx. Volume	Source
CICIDS2017	PortScan category	~158,000 flows	UNB / CIC (Canada)
UNSW-NB15	Reconnaissance category	~13,000 records	UNSW (Australia)

5.2. Input Variables (*Features*)

Variable	Type	Relevance	Role
Distinct Dst Ports / IP	Quantitative	Critical	Core scanning signature
SYN Flag Count	Quantitative	Critical	SYN stealth scan detection
RST Flag Count	Quantitative	Very High	Probe rejection indicator
Flow Duration	Quantitative	High	Very short in fast scans
Total Fwd Packets	Quantitative	High	Typically 1 per probe
IAT Mean	Quantitative	High	Large in slow scans
Port Range Entropy	Quantitative	High	Randomized vs sequential scan
ACK Flag Count	Quantitative	Medium	ACK scan detection
Unique Dst IPs / Src	Quantitative	Medium	Host discovery phase
Bwd Packet Length	Quantitative	Medium	RST vs SYN-ACK ratio
TTL Value	Quantitative	Low	OS fingerprinting attempts

5.3. Output Variable (*Label*)

Variable	Type	Values
Traffic class (v1)	Qualitative	BENIGN / PORT_SCAN
Scan type (v2)	Qualitative	SYN / UDP / ACK / XMAS / Connect